

**GANGA INSTITUTE OF TECHNOLOGY AND MANAGEMENT
JHAJJER, HARYANA**

**A PROJECT REPORT ON
PIR Motion Detection System Using Arduino UNO**

SUBMITTED OF THE REQUIREMENTS FOR THE TASK-2 ASSIGNMENT

MAINCRAFTS TECHNOLOGY INTERNSHIP:

INTRODUCTION TO EMBEDDED SYSTEM& IOT DEVICES DESIGN

**SUBMITTED TO :
MAINCRAFTS TECHNOLOGY**

**SUBMITTED BY:
NAME - AMAN KUMAR
COLLEGE ROLL -24ECE005
PROGRAM- INTRODUCTION
TO EMBEDDED SYSTEM &
IOT DEVICE DESIGN**

MOTION SENSOR PROJECT

INTRODUCTION TO EMBEDDED SYSTEM

What is an Embedded System

An embedded system is a computer-based system designed to perform a specific task. It consists of hardware (microcontroller, sensors, memory) and software (firmware) working together to control a device. Unlike a general-purpose computer, an embedded system is dedicated to one function and operates efficiently in real time.

Introduction

A Motion Detection System is an embedded system that uses a Passive Infrared (PIR) Sensor to detect the movement of humans or animals by sensing changes in infrared radiation. When motion is detected, the PIR sensor sends a signal to the Arduino UNO. The Arduino processes this signal, turns ON an LED, and displays a message on the Serial Monitor. This project demonstrates the fundamentals of sensor interfacing, embedded programming, and real-time hardware control.

Objective

The objective of this project is to design and develop a Motion Detection System using a PIR Sensor and Arduino UNO. The system is designed to detect human movement by sensing changes in infrared radiation. When motion is detected, the Arduino processes the sensor signal, turns on an LED, and displays the motion status on the Serial Monitor. This project aims to provide practical knowledge of sensor interfacing, embedded C/C++ programming, and real-time hardware control while demonstrating the basic concepts of automation and security systems

Working Principal

The working principle of this project is based on the Passive Infrared (PIR) Sensor, which detects changes in infrared radiation emitted by the human body or moving objects. The PIR sensor continuously monitors its surroundings and sends a HIGH signal to the Arduino UNO whenever motion is detected. The Arduino reads this signal through a digital input pin and immediately turns ON the LED while displaying "Motion Detected!" on the Serial Monitor. When no motion is detected, the PIR sensor sends a LOW signal, causing the Arduino to turn OFF the LED and display "No Motion" on the Serial Monitor.

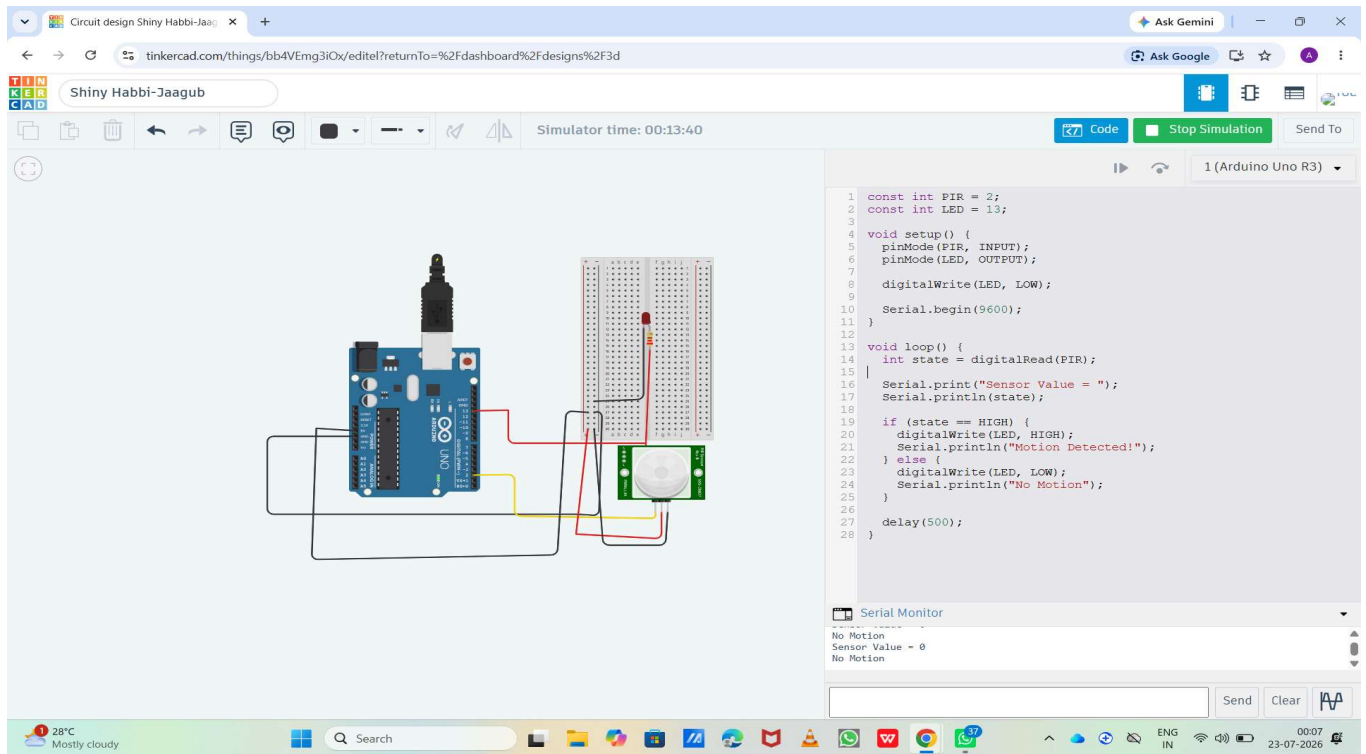
Required Components

- **Arduino Uno**
- **PIR Motion Sensor**
- **LED**
- **220Ω Resistor**
- **Breadboard**
- **Jumper Wires**

Component Description

- **Arduino UNO** - The Arduino UNO is the main microcontroller board used in this project.
- **PIR Motion Sensor (HC-SR501)** - The PIR (Passive Infrared) Motion Sensor detects changes in infrared radiation emitted by the human body.
- **LED (Light Emitting Diode)** - The LED acts as a visual indicator. It turns ON when motion is detected and turns OFF when no motion is detected.
- **220Ω Resistor** - The 220-ohm resistor limits the current flowing through the LED, protecting it from damage and ensuring safe operation.
- **Breadboard** - The breadboard is used to build the circuit without soldering. It provides an easy and organized way to connect electronic components.
- **Jumper Wires** - Jumper wires are used to establish electrical connections between the Arduino UNO, PIR sensor, LED, resistor, and breadboard

Circuit Diagram



Programming

```
const int pirPin = 2;
const int ledPin = 13;

void setup() {
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int pirState = digitalRead(pirPin);

  if (pirState == HIGH) {
    digitalWrite(ledPin, HIGH);
    Serial.println("Motion Detected!");
  } else {
    digitalWrite(ledPin, LOW);
    Serial.println("No Motion");
  }

  delay(100);
}
```

Real world applications

- **Home Security Systems** - PIR sensors are widely used in home security systems to detect unauthorized movement.
- **Smart Lighting Systems** - PIR sensors automatically turn lights ON when a person enters a room and switch them OFF when no motion is detected, helping to save electricity and improve energy efficiency.
- **Office and Building Security** - These sensors are installed in offices, schools, banks, and commercial buildings to monitor movement in restricted areas and enhance security.
- **Automatic Door Systems** - PIR sensors are used in automatic sliding doors at shopping malls, hospitals, airports, and offices.
- **Warehouse and Factory Monitoring** - Industries use PIR sensors to monitor movement in warehouses, storage rooms, and production areas, improving safety and reducing unauthorized access.